

Let, $(P(1) \wedge \neg P(2) \wedge \neg P(3)) \vee (P(2) \wedge \neg P(1) \wedge \neg P(3)) \vee (P(3) \wedge \neg P(1) \wedge \neg P(2))$ be True. By defn. of OR, atleast one of $(P(1) \wedge \neg P(2) \wedge \neg P(3))$, $(P(2) \wedge \neg P(1) \wedge \neg P(3))$, $(P(3) \wedge \neg P(1) \wedge \neg P(2))$ is True.

Let, $P(1) \wedge \neg P(2) \wedge \neg P(3)$ be True. By defn. of AND, $P(1)$ is True, $\neg P(2)$ is True and $\neg P(3)$ is True. By defn. of negation, $\neg(\neg P(2))$ and $\neg(\neg P(3))$ are both False. By double negation law, $P(2)$ and $P(3)$ are both False. By defn. of uniqueness quantifier, $\exists! x P(x)$ is True.

By defn. of negation, $\neg P(1)$ is False. By defn. of AND, $(P(2) \wedge \neg P(1) \wedge \neg P(3))$ and $(P(3) \wedge \neg P(1) \wedge \neg P(2))$ are both False.

Let, $P(2) \wedge \neg P(1) \wedge \neg P(3)$ is True. By defn. of AND, $P(2)$ is True, $\neg P(1)$ is True and $\neg P(3)$ is True. By defn. of negation, $\neg(\neg P(1))$ and $\neg(\neg P(3))$ are both False. By double negation Law, $P(1)$ and $P(3)$ are both False. By defn. of uniqueness quantifier, $\exists! x P(x)$ is True.

By defn. of negation, $\neg P(2)$ is False. By defn. of AND, $(P(1) \wedge \neg P(2) \wedge \neg P(3))$ and $(P(3) \wedge \neg P(1) \wedge \neg P(2))$ are both False.

Let, $P(3) \wedge \neg P(1) \wedge \neg P(2)$ is True. By defn. of AND, $P(3)$ is True, $\neg P(1)$ is True, $\neg P(2)$ is True. By defn. of negation, $\neg(\neg P(1))$ and $\neg(\neg P(2))$ are both False. By ~~defn.~~ double negation law, $P(1)$ and $P(2)$ are both False. By defn. of uniqueness quantifier, $\exists! x P(x)$ is True.

By defn. of negation, $\neg P(3)$ is False. By defn. of AND, $(P(1) \wedge \neg P(2) \wedge \neg P(3))$ and $(P(2) \wedge \neg P(1) \wedge \neg P(3))$ are both False.

$\therefore$ Exactly one of $(P(1) \wedge \neg P(2) \wedge \neg P(3))$, $(P(2) \wedge \neg P(1) \wedge \neg P(3))$ or $(P(3) \wedge \neg P(1) \wedge \neg P(2))$ is True at any time.

$(P(1) \wedge \neg P(2) \wedge \neg P(3)) \vee (P(2) \wedge \neg P(1) \wedge \neg P(3)) \vee (P(3) \wedge \neg P(1) \wedge \neg P(2))$ is True
 $\Rightarrow \exists! x P(x)$ is True

They are logically equivalent $\square$